

ASPECT Pipeline Data User Manual

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Applicable Documents

AD	Document Title
AD-01	Hera Image Format and Calibration Levels
AD-02	PFM and FS default wl and setpoints.xlsx

Reference Documents

[RD1] M. Cardi et al. “The Hera Milani Mission”. In: *Space Science Reviews* (2025). under review.

[RD2] Lucio Chiappetti et al. *Definition of the Flexible Image Transport System (FITS): The FITS Standard Version 4.0*. Tech. rep. Accessed: 2024-12-20. NASA/IAU, 2018. URL: https://fits.gsfc.nasa.gov/fits_standard.html.

[RD3] SemiConductor Devices. *SWIR640 FPA User Guide*. 2020.

[RD4] P. Michel et al. “The Hera Space Mission in the Context of Small Near-Earth Asteroid Missions in the Past, Present and Future”. In: *Space Science Reviews* 221.70 (2025). DOI: 10.1007/s11214-025-01195-1.

[RD5] A. Näsilä et al. “The Asteroid Spectral Imager (ASPECT) on the Milani CubeSat”. In: *Space Science Reviews* (2025). accepted.

Acronyms

ASPECT	Asteroid Spectral Imager
CDS	Correlated double sampling
DN	Digital Number
FITS	Flexible Image Transport System
FPI	Fabry-Perót Interferometer
HDU	Header and Data Unit
MGM	Modified Gaussian Model
NIR	Near infrared (wavelengths)
SWIR	Short-wave infrared (wavelengths)
Vis	Visible (wavelengths)

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1. Introduction

ASPECT (Asteroid Spectral Imager) [RD5] is a hyperspectral camera on board of the Milani CubeSat [RD1] of the ESA Hera mission [RD4]. Aspect consists of three independent measurement channels - visible (Vis), near infrared 1 and 2 (NIR 1 and NIR 2), and a short wave infrared (SWIR). The instrument parameters are specified in Table 1 [RD5]. ASPECT purpose is to map the surface maturity and mineral composition of the target binary asteroid Didymos/Dimorphos. The ASPECT data processing pipeline was developed to facilitate an accurate ground calibration of the acquired instrument data, fuse observations from all four sensors, and turn these into scientific data products through the application of advanced analysis algorithms. This document is intended for end users of ASPECT data products. It provides a clear description of each pipeline stage, outlining the structure and expected content formats of the resulting data products.

Table 1: ASPECT imager parameters [RD5].

Parameter	Vis channel	NIR1 channel	NIR2 channel	SWIR channel
Field of view (deg)	$10^{\circ} \times 10^{\circ}$	$6.7^{\circ} \times 5.4^{\circ}$	$6.7^{\circ} \times 5.4^{\circ}$	5.85° circular
Spectral range (nm)	650–900	850–1250	1200–1600	1650–2500
Image size (px)	1024×1024	640×512	640×512	1 pixel
No. spectral bands	14	14	14	30
Spectral resolution (nm)	< 30	< 40	< 40	< 45

2. Overview of pipeline steps

The ASPECT pipeline is organized into distinct processing levels, each corresponding to a specific stage of the calibration or analysis. Each processing level consist of one or more steps. Beginning with raw data, each level progressively applies data calibration or processing procedures. Each processing level outputs its results in an FITS file. The naming convention of the data files and the structure of each processing level follow the guidelines defined in [AD-01] so the imaging channel and processing level can be easily identified from the FITS file name. Characters 1-4 specify the ASPECT channel data included: ASx_ refers to a specific channel while ASP_ denotes a product combining multiple channels, where x indicates the channel index—0 for Vis, 1 for NIR1, 2 for NIR2, and 3 for SWIR. Characters 26-27 of the filename indicate the processing level. For example, ASP_009FEQ_270323T060000_2B.fits refers to a data product combining multiple channels, processed to Level 2B.

The Figure 1 outlines processing levels applicable to each channel. Columns represents the individual imaging channels (Vis, NIR1, NIR2, or SWIR) while the rows correspond to a specific processing steps associated with a particular processing levels. A solid square within the diagram indicates that the corresponding step of the respective level is applicable to a particular channel. Each processing level builds upon the output of the preceding level. The level 3, which represents the data analysis stage, is applied to a combined hyperspectral cubes that containing data from one or more spectral channels.

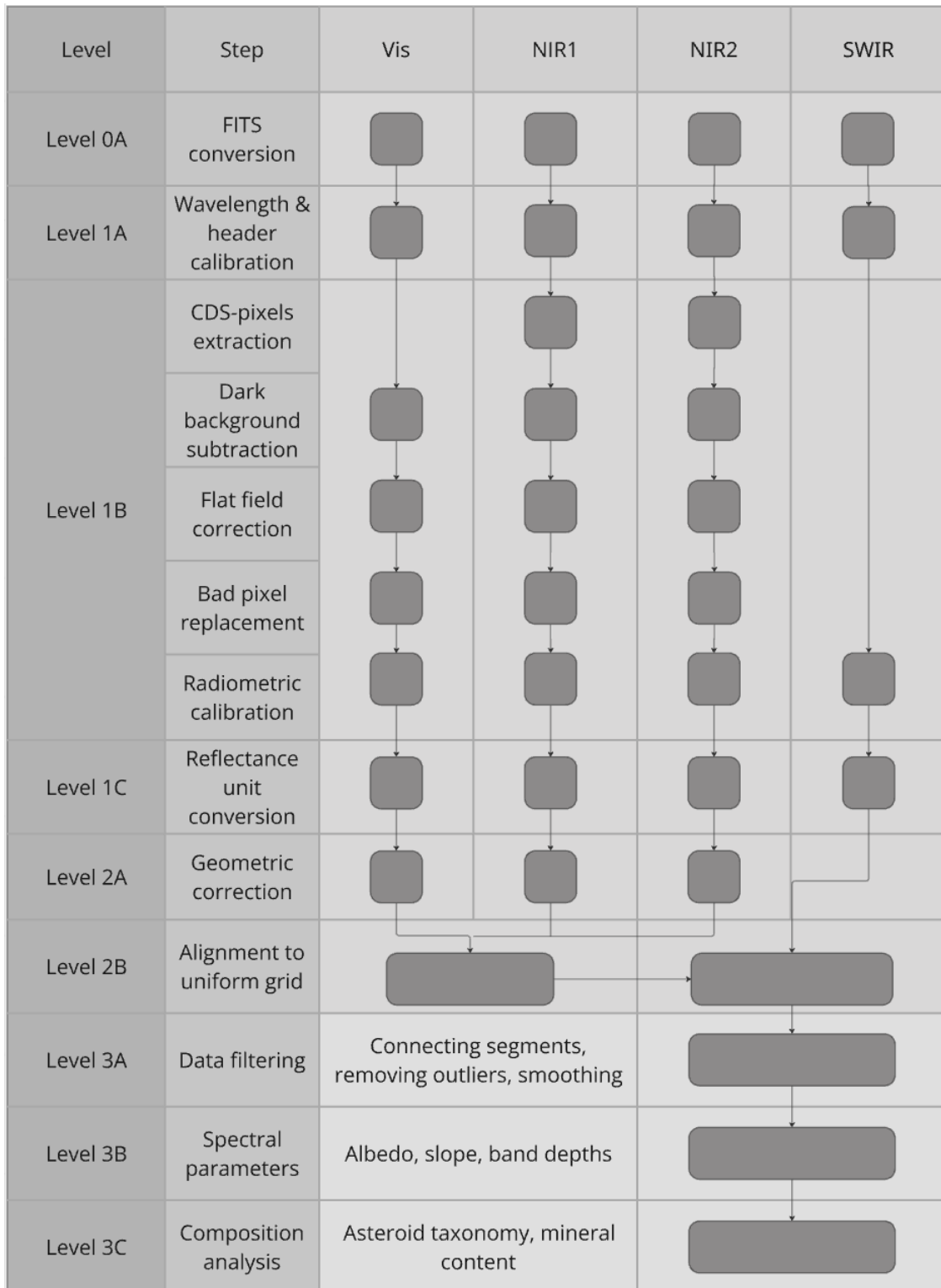


Figure 1: The ASPECT pipeline flow diagram.

3. Data products

This section describes the overall structure of data products produced by the pipeline. All FITS files created by the pipeline conform to the FITS standard version 4.0, as defined in [RD2].

Each FITS file produced by the pipeline contains a Primary HDU, which holds all relevant meta-data, and a data unit containing either a three-dimensional hyperspectral data cube (for Vis, NIR1, and NIR2 channels) or an one-dimensional array (for SWIR) derived from the corresponding raw acquisition binary files.

For NIR1 and NIR2 channels, an additional Binary Table extension is included from processing level 1B onward. This extension stores the CDS pixels extracted from the original near-infrared images [RD3].

The level 1 FITS file structure for each channel is depicted in the Figure 2. The FITS structures corresponding to Levels 2 and 3 are described in dedicated sections.

FITS structure	Vis	NIR1	NIR2	SWIR
Primary HDU	✓	✓	✓	✓
Header	Acquisition metadata	Acquisition metadata	Acquisition metadata	Acquisition metadata
Data unit	Image cube	Image cube	Image cube	Array
Binary Table HDU	✗	✗ ✓	✗ ✓	✗
Header		Only standard	Only standard	
Data unit		CDS-pixels after level 1B	CDS-pixels after level 1B	

Figure 2: FITS file structure by instrument channel.

3.1. Primary HDU

The Primary HDU contains all metadata related to the image acquisition. This includes the information about the acquisition, relevant SPICE metadata, calibration-specific metadata, and the instrument channel specific metadata. In accordance with the FITS standard, the Primary header begins with a set of mandatory default entries, followed by the acquisition metadata entries specified in the Table 2.

The instrument-specific metadata entries are indexed according to the same channel mapping used in the file naming convention described above. For example, the header field '0_WL' contains the wavelength values for the Vis channel, while '1_WL' contains those of the NIR1 channel. Similarly, entries prefixed with '2_' and '3_' corresponds to the NIR2 and SWIR channels, respectively.

The instrument-specific include the key temperature readings. The CCDTMP (detector temperature), FPI1 (spectral filter temperature 1), and FPI2 (spectral filter temperature 2) are provided for each channel, even if no other data from that channel is included in the data product. Other instrument-specific metadata entries, including FRAMES, ORDER, EXPOS, SP1, SP2, SP3, and WL, are present only if the corresponding measurement data for a particular channel are included in the FITS file. The channels present in the data product are listed as a comma-separated string in the 'CHANNELS' header entry.

Table 2: Primary header metadata entries.

Key	Description	Type
INSTRUME	Camera ID, 'ASPECT'	String
ORIGIN	'ESA-HERA'	String
DATE	UTC time of file creation, e.g. '2025-07-31T15:13:47'	String
FILENAME	Name of the FITS file	String
SWCREATE	Software identification, e.g. 'ASPECTCAL v1.0'	String

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Table 2: Primary header metadata entries.

Key	Description	Type
ORIGFILE	Original filename, e.g. 'dc_X_exp_XXX.bin'	String
PROCLEVEL	Current calibration level, same as at the end of the filename	String
MISSPHAS	HERA Mission Phase ID	String
OBSERVPH	HERA Observation ID	String
OBSTARGET	Observation target, SPICE format	String
DATE-OB	UTC time of observations, e.g. '2027-03-23T06:00:00.000'	String
OBJECT	Observation object	String
SC_CLK	Spacecraft clock Hera instrument format	String
SPICE_MK	SPICE metakernel	String
SPICEVER	SPICE dataset version	String
SPICECLK	SC clock SPICE format, e.g. '1/077436069.1856'	String
SUN_POSX	Sun position vector X [km]	Float
SUN_POSY	Sun position vector Y [km]	Float
SUN_POSZ	Sun position vector Z [km]	Float
SOLAR_D	Solar distance [AU]	Float
EARTPOSX	Earth position vector X [km]	Float
EARTPOSY	Earth position vector Y [km]	Float
EARTPOSZ	Earth position vector Z [km]	Float
EARTH_D	Earth distance [AU]	Float
TARGET	Observation target (SPICE)	String
TRG_POSX	Target position vector X [km]	Float
TRG_POSY	Target position vector Y [km]	Float
TRG_POSZ	Target position vector Z [km]	Float

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Table 2: Primary header metadata entries.

Key	Description	Type
TRG_DIST	Target distance [au]	Float
SC_QUATW	Spacecraft quaternion (W)	Float
SC_QUATX	Spacecraft quaternion (X)	Float
SC_QUATY	Spacecraft quaternion (Y)	Float
SC_QUATZ	Spacecraft quaternion (Z)	Float
CAM_RA	Camera axis RA [deg]	Float
CAM_DEC	Camera axis DEC [deg]	Float
CAM_NAZ	Camera axis north azimuth [deg]	Float
SOL_ELNG	Solar elongation	Float
CALPHASE	Calibration phase	String
CHANNELS	Channel data in this file, e.g. ('Vis,NIR1,NIR2')	String
0_CCDTMP	Detector temperature	String
0_FPI1	FPI 1 temperature	String
0_FPI2	FPI 2 temperature	String
0_FRAMES	Frame indexes included, comma-separated if multiple	String
0_ORDER	LOW or HIGH	String
0_EXPOS	Exposure time, comma-separated if multiple	String
0_SP1	Piezo setpoint 1, comma-separated if multiple	String
0_SP2	Piezo setpoint 2, comma-separated if multiple	String
0_SP3	Piezo setpoint 3, comma-separated if multiple	String
0_WL	Wavelengths after 1A, comma-separated if multiple	String
1_CCDTMP	Detector temperature	String
1_FPI1	FPI 1 temperature	String

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Table 2: Primary header metadata entries.

Key	Description	Type
1_FPI2	FPI 2 temperature	String
1_FRAMES	Frame indexes included, comma-separated if multiple	String
1_ORDER	LOW or HIGH FPI interference order	String
1_EXPOS	Exposure time, comma-separated if multiple	String
1_SP1	Piezo setpoint 1, comma-separated if multiple	String
1_SP2	Piezo setpoint 2, comma-separated if multipl.	String
1_SP3	Piezo setpoint 3, comma-separated if multiple	String
1_WL	Wavelengths after 1A, comma-separated if multiple	String
2_CCDTMP	Detector temperature	String
2_FPI1	FPI 1 temperature	String
2_FPI2	FPI 2 temperature	String
2_FRAMES	Frame indexes included, comma-separated if multiple	String
2_ORDER	LOW or HIGH FPI interference order	String
2_EXPOS	Exposure time, Comma-separated if multiple.	String
2_SP1	Piezo setpoint 1, Comma-separated if multiple.	String
2_SP2	Piezo setpoint 2, Comma-separated if multiple.	String
2_SP3	Piezo setpoint 3, Comma-separated if multiple.	String
2_WL	Wavelengths after 1A, comma-separated if multiple.	String
3_CCDTMP	Detector temperature	String
3_FPI1	FPI 1 temperature	String
3_FPI2	FPI 2 temperature	String
3_FRAMES	Frame indexes included, comma-separated if multiple	String
3_ORDER	LOW or HIGH FPI interference order	String

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Table 2: Primary header metadata entries.

Key	Description	Type
3_EXPOS	Exposure time, comma-separated if multiple	String
3_SP1	Piezo setpoint 1, comma-separated if multiple	String
3_SP2	Piezo setpoint 2, comma-separated if multiple	String
3_SP3	Piezo setpoint 3, comma-separated if multiple	String
3_WL	Wavelengths after 1A, comma-separated if multiple	String

3.2. Binary Table HDU

Following the Level 1B processing the FITS files containing NIR1 or NIR2 data will include an additional Binary Table extension. This extension stores the CDS pixels that are extracted from each NIR frame. In the original NIR1 and NIR2 images, the first six rows, the last row, and the four outermost columns on both sides consist of CDS pixels that do not form part of the actual image content. These pixels are removed during Level 1B processing and stored in the Binary Table. Each column in the table corresponds to a single frame and contains a one-dimensional array of 7,984 values, where the pixels are stored in row-major order, with pixels within each row listed from left to right. The column name encodes the channel index and frame number. For example, the column labeled 1_000 contains the CDS-pixels for frame 000 of the NIR1 channel. In FITS files combining multiple channels the CDS pixel tables from individual NIR1 and NIR2 frames are combined into a single Binary Table.

4. Pipeline levels

This section provides a detailed overview of how each processing level transforms the ASPECT data. For each level, the structure and content of the resulting data are described, including changes to the image data format, the introductions of additional data extensions, and updates to the relevant FITS header metadata. Each processing level will modify at least the header fields 'FILENAME' and 'PROCLEVEL' to match the current level. The format and data type of the data vary depending on the channel characteristics and the processing level applied as illustrated in the Figure 3. The amount of the image frames per instrument channel, denoted by n in the table, may also vary. In the default ASPECT acquisition sequence defined in [AD-02], n is 11 for Vis, 13 for NIR1, 13 for NIR2, and 23 for SWIR.

Processing level	Vis	NIR1	NIR2	SWIR
0A	($n \times 1024 \times 1024$) uint16	($n \times 518 \times 648$) uint16	($n \times 518 \times 648$) uint16	(n ,) uint32
1A	($n \times 1024 \times 1024$) uint16	($n \times 518 \times 648$) uint16	($n \times 518 \times 648$) uint16	(n ,) uint32
1B	($n \times 1024 \times 1024$) float32 (>f4)	($n \times 512 \times 640$) float32 (>f4)	($n \times 512 \times 640$) float32 (>f4)	(n ,) float32 (>f4)
1C	($n \times 1024 \times 1024$) float32 (>f4)	($n \times 512 \times 640$) float32 (>f4)	($n \times 512 \times 640$) float32 (>f4)	(n ,) float32 (>f4)
2B	(n x 512 x 640) float32 (>f4)			(n,) float32 (>f4)
3	TBD			

Figure 3: The ASPECT Data Unit shape and data type by processing level.

4.1. Level 0A

The purpose of level 0A is to convert the original raw binary images of a given channel into a single FITS file containing the hyperspectral data cube of one acquisition. The resulting FITS file

consists of a Primary HDU with header entries as defined in Table 2, containing raw uncalibrated DN values. With a few exceptions described in Section 3.1, only instrument temperature values are included from other channels. In addition, the wavelength field ('x_WL', where 'x' denotes the instrument channel ID) is only added at Level 1A. The image data is stored in Primary Data Unit in its original, uncalibrated form.

4.2. Level 1A

Level 1A processing affects only the Primary Header of the level 0A data. Its purpose is to convert instrument temperature readings from raw DN values to calibrated temperatures in Kelvin and to convert the piezo setpoint values into actual wavelength values. The header fields updated at this stage are listed in Table 3.

Table 3: Header fields modified at Level 1A.

Key	Description	Type
FILENAME	Characters 26-27 updated to match the processing level	String
PROCLEVEL	Processing level identifier updated. e.g. '1A'	String
x_CCDTMP	Values are converted from DNs to Kelvins (and Celsius)	String
x_FPI1	Values are converted from DNs to Kelvins (and Celsius)	String
x_FPI2	Values are converted from DNs to Kelvins (and Celsius)	String
x_EXPOS	Values are converted from DNs to Kelvins (and Celsius)	String
x_WL	Channel x wavelengths converted from x_SP1 values	String

4.3. Level 1B

Level 1B processing modifies the data cube by calibrating the raw pixel values into radiance values in $\text{W m}^{-2} \text{sr}^{-1} \text{nm}^{-1}$ unit. The applied calibration steps are described in Table 4. The header the field 'CALPHASE' will contain a comma-separated list (as a string) of all the calibration steps applied to the data.

Table 4: Calibration steps at Level 1B.

Step	Description
Float64 conversion	The pixel values are converted to double precision for calculations
CDS-pixel extraction	Extracts CDS pixels from NIR1 and NIR2 are extracted and appended to Binary Table extension
Dark background subtraction	Remove dark current based on the master dark frame
Flat field correction	Flat field correction
Bad pixels replacement	Replace bad pixels by neighboring pixels median
Radiometric calibration	Scale the image pixels to radiance by the absolute calibration factor
Float32 conversion	Convert pixel values to float32 for storing

4.4. Level 1C

Level 1C modifies the result of level 1B by pixel linear transformations. In the resulting file the pixel values are represented as dimensionless I/f reflectance units.

4.5. Level 2A

Level 2A modifies the data by resampling the 1C image by applying the distortion correction algorithm.

4.6. Level 2B

Level 2B merges the selected instrument channels into a single FITS file containing all frames from the combined channels. The included channels are recorded in the 'CHANNELS' header field as a comma-separated string. The resulting Primary Header follows the structure defined in Table 2.

When Vis data is combined with NIR channels, the Vis images are aligned and resamples to match the NIR channel dimensions (512, 640) using the pipeline's alignment and resampling algorithm. If the SWIR channel is included, its data is stored as a one-dimensional array in an Image HDU placed immediately after the Primary HDU. CDS pixels from NIR1 and NIR2 (if present) are appended as a single Binary Table extension at the end of the FITS file.

4.7. Level 3

Level 3 is the data analysis stage of the pipeline. It processes the output from Level 2B using algorithms such as MGM or a convolution neural network for spectral, composition, and taxonomy analyses. The results are stored in FITS files containing only a Primary HDU, which retains the same header content as in Level 2B, with the algorithm output stored in the Primary Data Unit. For the analyses, single-pixel asteroid spectra are first extracted from the data cube. Pixels corresponding to the asteroid are identified, and then spectral profiles across the cube are retrieved for these locations.

The outputs are then assembled into a two-dimensional images of size 512 × 640, where each pixel corresponding to an extracted asteroid location contains a specific scientific analytical parameter for the associated spectrum, and all remaining pixels (asteroid background) are set to zero.